

Durham College
Network Monitoring and Penetration Testing
INFT 1202

Lab 2: Install Zabbix SNMP Agent on a Windows Server 2019

Lab Workbook

Name – Student ID	Course Code	Weight
100899671	INFT1202	5%

Part 1: Windows 2019 Server Installation

Part 2: Heavy loader application installation on Windows server

Part 3: Enable Zabbix monitoring on Windows 2019 server

Part 4: Execute the load runner on Windows to generate load

Lab Objectives

- Install Windows 2019 Virtual Machine (VM).
- Monitor the Windows server 2019 server using Zabbix
- Run Heavy loader application on Windows server 2019

Lab Materials

- Please name your Lab workbook titled “Lab 2 - <name> <Student ID>”.
 - Please include all relevant screenshots on Lab workbook for submission (provided on dc connect)
- Install Windows Server 2019 VM
- Use Zabbix to monitor the Windows Server 2019

Lab Requirements

- VMware workstation 16 pro
- Windows server 2019
- Zabbix installation from Lab 1
- Heavy loader application

Lab 2 Week 3 – Zabbix SNMP Agent Installation

PART 1: WINDOWS 2019 SERVER INSTALLATION

1. Go to your Azure cloud portal “<https://portal.azure.com/>” to download windows server 2019 **(If you have a previously installed Windows 2019/2022 server, you can clone that for this task)**
2. Download windows server 2019/2022.

Software

Education



Windows Server 2019 Datacenter (updated Sept 2019)

Windows Server 2019 Standard is ideal for customers with low density or non-virtualized environments. Windows Server 2019 offers additional features in Standard and Datacenter editions. Features exclusive to the Windows Server 2016 Datacenter edition include Shielded Virtual Machines, software-defined networking, Storage Spaces Direct, and Storage Replica. While no features from the Windows Server 2016 R2 Standard edition have been removed, we have added features like Nano Server and unlimited Windows Server containers to the Windows Server 2019 Standard edition.

Operating System

Windows

Product language

English

System

64 bit

[View Key](#)

3. Install the windows 2019 server on your VMware Workstation (I believe you all should know how to install this by now)

INFT 1202

Microsoft Azure Education | Software

Software	Category	Operating System	Architecture
Machine Learning Server for Windows	AI + Machine Learning	Windows	64 bit
Microsoft Hyper-V Server 2019 (updated Sept 2019)	Compute	Windows	64 bit
Microsoft R Server 9.1.0 for Hadoop	Database	Windows	64 bit
Microsoft R Server 9.1.0 for Linux	Database	Linux	64 bit
Microsoft R Server 9.1.0 for Teradata	Database	Windows	64 bit
Microsoft R Server 9.1.0 for Windows	Database	Windows	64 bit
SharePoint Server Subscription Edition 2019	Productivity Tools	Windows	64 bit
SharePoint Server Subscription Edition 2019	Productivity Tools	Windows	64 bit
Skype for Business Server 2019	Productivity Tools	Windows	64 bit
Windows Server 2022 Datacenter	Operating System	Windows	64 bit
Windows Server 2019 Datacenter (updated Sept 2019)	Operating System	Windows	64 bit
Windows Server 2019 Essentials (updated Sept 2019)	Operating System	Windows	64 bit
Windows Server 2019 Features on Demand	Operating System	Windows	64 bit
Windows Server 2019 Language Pack	Operating System	Windows	64 bit
Windows Server 2022 Standard	Operating System	Windows	64 bit
Windows Server 2019 Standard (updated Sept 2019)	Operating System	Windows	64 bit
SQL Server 2019 Standard	Database	Windows	64 bit

Software
Education

Windows Server 2019 Datacenter (updated Sept 2019)

Windows Server 2019 Standard is ideal for customers with low density or non-virtualized environments. Windows Server 2019 offers additional features in Standard and Datacenter editions. Features exclusive to the Windows Server 2016 Datacenter edition include Shielded Virtual Machines, software-defined networking, Storage Spaces Direct, and Storage Replica. While no features from the Windows Server 2016 R2 Standard edition have been removed, we have added features like Nano Server and unlimited Windows Server containers to the Windows Server 2019 Standard edition.

Operating System
Windows

Product language
English

System
64 bit

[View Key](#)

[Help improve this page](#)

- After the installation, Go to VM > Settings of the new installed Windows server virtual machine to change the memory and processor settings.
 - Memory: 4 GB
 - Processors: 1

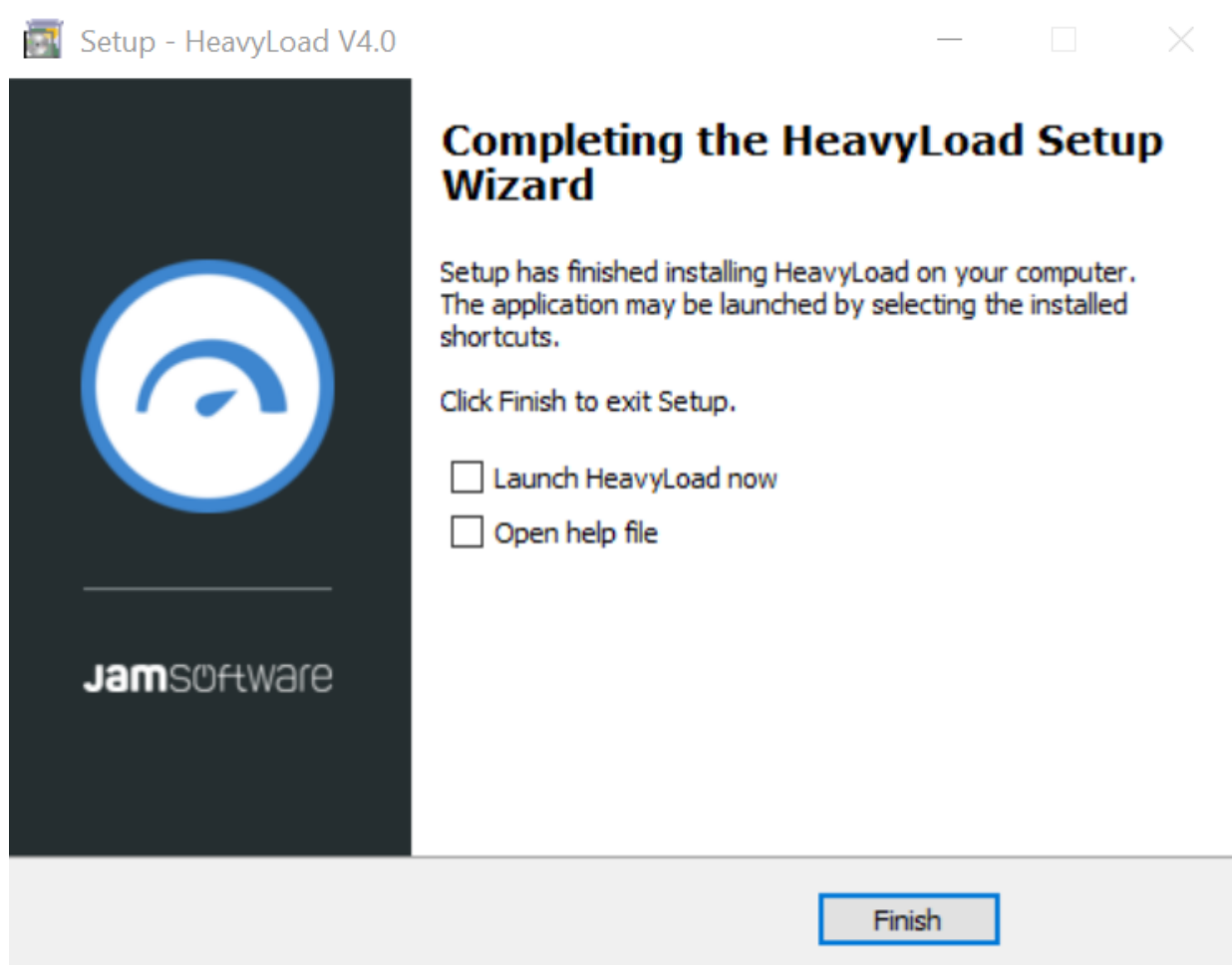
INFT 1202

PART 2: HEAVY LOADER APPLICATION INSTALLATION ON WINDOWS SERVER

Turn on your newly installed Windows server to install the heavy loader application.

1. Go to the website <https://www.jam-software.com/heavyload?ca=1> to download the heavy loader application setup.
2. This can be downloaded on your desktop or download folder. After download, double click on the setup to install.
3. Click yes to allow the application to run and follow the prompt
4. You can choose to create a desktop shortcut after installation.
5. Uncheck the Launch heavy Load now and click finish

[Attach Screenshot]



INFT 1202

PART 3: ENABLE ZABBIX MONITORING ON WINDOWS 2019 SERVER

Go to the windows server 2019 and install SNMP using the server manager

1. Install SNMP Service in Windows Server 2019.
 - Open Server Manager
 - Click Add roles and features
 - Under Before you begin click Next
 - Under Select installation type click Next
 - Under Select destination server click Next
 - Under Select server roles click Next
 - Under Select features select SNMP Service and click Add Features and then click Next
2. Select SNMP Service
 - Under Confirm installation selection click, Install
 - Wait until the installation is finished
3. Once the installation is done, Open the Services applet (click on the Start menu and search for services) and click Apply and OK to start the 'SNMP Service'
4. Click the 'Agent' tab and specify the following details:
 - Contact: ZabbixMonitor
 - Location: Windows-SERVER
 - Here, select the following services:
 - Physical, Applications, Internet, End-to-end, then click 'OK'.

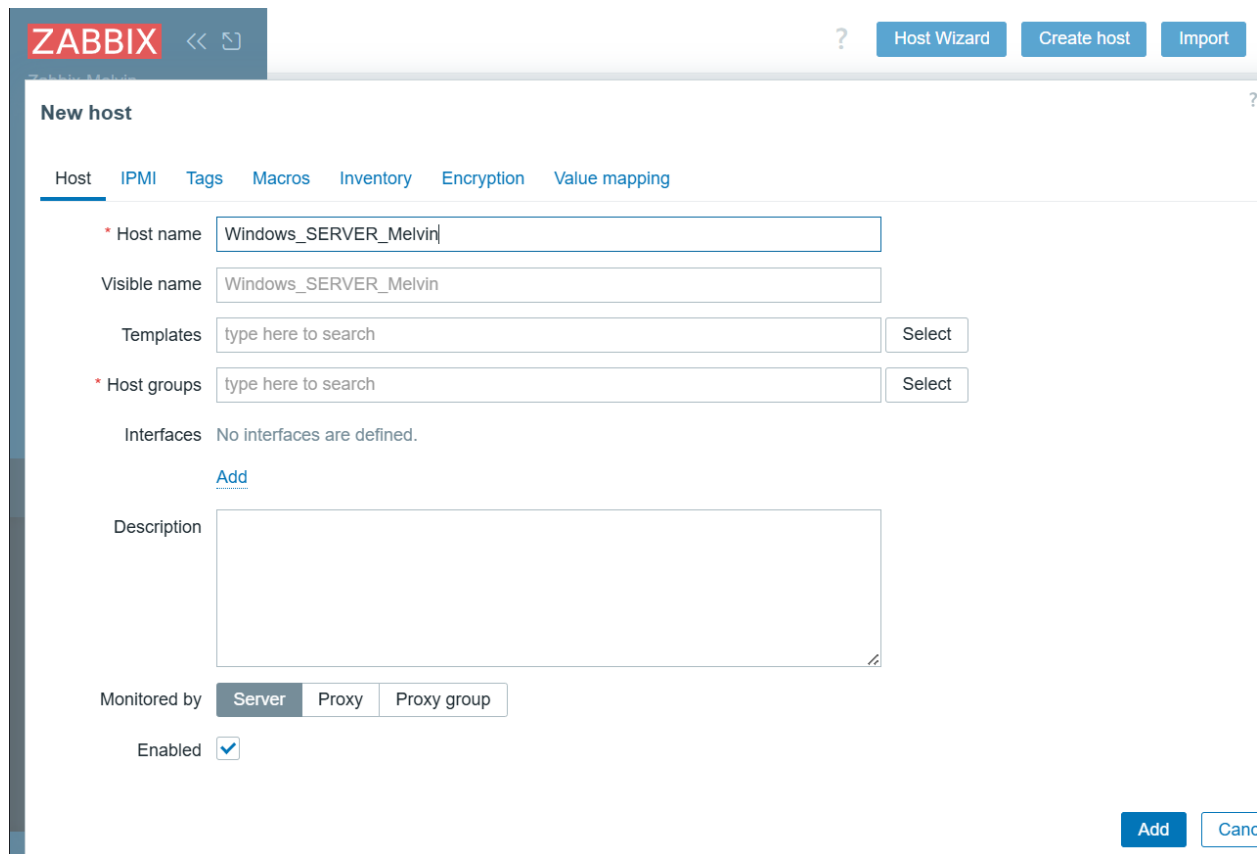
INFT 1202

- Click the 'Security' tab on the same dialog box and click Add, to add the community.
- Input the community name is 'ZabbixMonitor' here and the Community rights: 'READ ONLY' then click 'Add'.
- Select 'Accept SNMP packets from any host'
- The next step would be to add your Windows server Machine as a host in your Zabbix instance.

Go to the 'Data Collection' tab in your Zabbix, then click 'Hosts'.

- Click 'Create host'.
- Use the host name: Windows_SERVER_Yourname, in my own case, I used Windows_SERVER_Fuad.

[Attach Screenshot]



ZABBIX << >> ? Host Wizard Create host Import

New host ?

Host IPMI Tags Macros Inventory Encryption Value mapping

* Host name

Visible name

Templates

* Host groups

Interfaces No interfaces are defined.
[Add](#)

Description

Monitored by ☒ Server ☐ Proxy ☐ Proxy group

Enabled ☒

- Then click on select by the templates section
- Click 'Select' to add the Template group as 'Templates/Operating Systems'.
- Choose 'Windows SNMP' and click 'Select'
- Now click the 'Select' in *Host groups.
- In Host groups, choose everything and click 'Select'.
- In Interfaces, click 'Add' to configure the SNMP Interface.
- Select SNMP from the drop down.
- Configure the following values for SNMP Interface:
 - IP address: (Your Windows Server 2019 IP Address) mine was 192.168.108.132
 - Port: 161

Fuad Mustapha
Fuad.mustapha@durhamcollege.ca

INFT 1202

- SNMP version: SNMPv2
- SNMP community: {\$SNMP_COMMUNITY}

Go to 'Macros' tab then enter the following details:

- Macro: {\$SNMP_COMMUNITY}
- Value: ZabbixMonitor

After Editing

[Attach Screenshot]

New host ?

Host IPMI Tags **Macros 1** Inventory Encryption Value mapping

Host macros Inherited and host macros

Macro	Value	Description
{\$SNMP_COMMUNITY}	ZabbixMonitor	description

[Add](#) [Remove](#)

[Add](#) [Cancel](#)

Then click 'Add'.

19. Confirm that the status of your newly added agent is enabled.

[Attach Screenshot]

Host added

Name	Items	Triggers	Graphs	Discovery	Web	Interface	Proxy	Templates	Status	Availability	Agent encryption	Info	Tags
Windows_SERVER_Melvin	Items 15	Triggers 7	Graphs 1	Discovery 3	Web	192.168.146.133:161		Windows by SNMP	Enabled	SNMP	None		
Zabbix server	Items 153	Triggers 83	Graphs 16	Discovery 6	Web	127.0.0.1:10050		Linux by Zabbix agent, Zabbix server health	Enabled	ZBX	None		

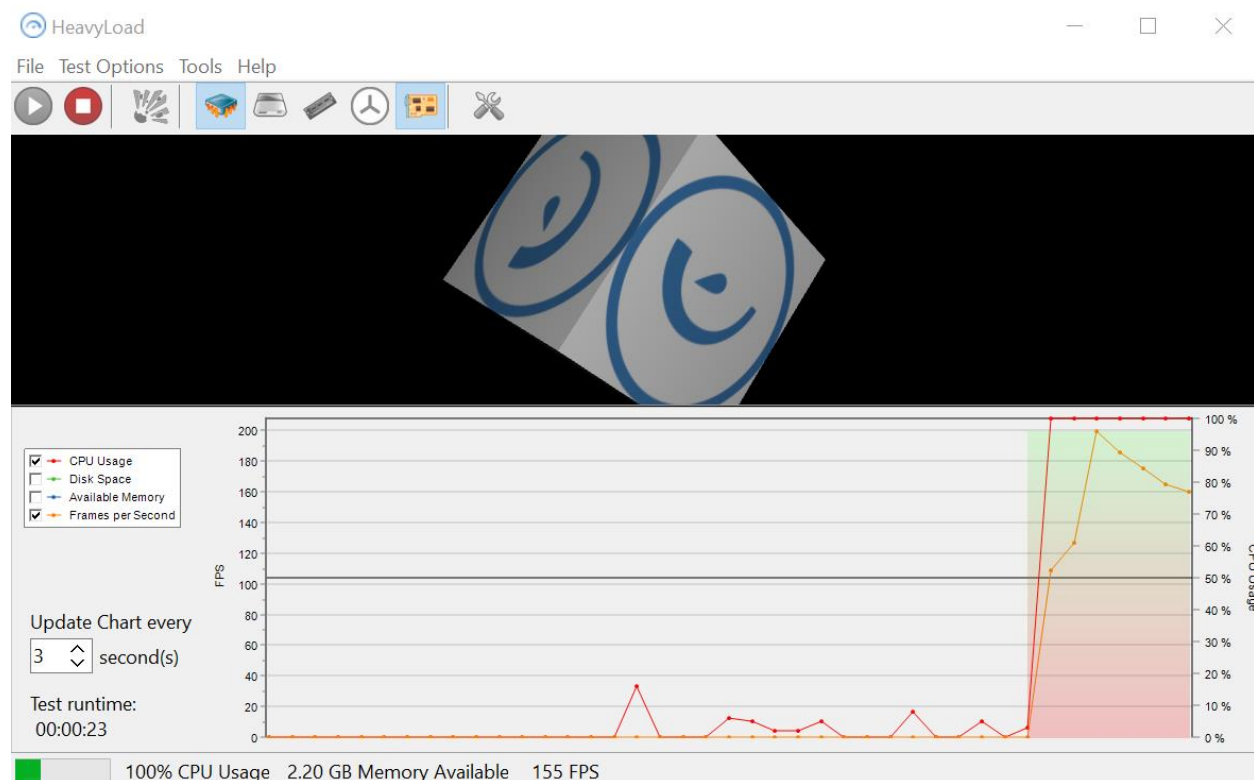
Displaying 2 of 2 found

INFT 1202

PART 4: EXECUTE THE LOAD RUNNER ON WINDOWS TO GENERATE LOAD

1. Launch the LoadRunner application from your windows server machine to generate a high CP usage.
2. This would simulate generating load on the Windows Machine and check the status in Zabbix.
3. Else manually run some applications on Windows-SERVER to generate load.

[Attach Screenshot]



4. Go over to your Zabbix instance through your browser and click on Dashboard, you should see a current problems tab. Showing a high memory utilization has been reported on your Zabbix server.

[Attach Screenshot]

Current problems								
Time ▼	Info	Host	Problem • Severity	Duration	Update	Actions	Tags	
12:52:21 AM		Windows_SERVER_Melvin	Windows: High CPU utilization (over 90% for 5m)	32s	Update		class: os component: cpu scope: performance	...

5. Click on the **Monitoring > hosts** in the left menu tab section to see the problems row. Click on the problems to see the details of the problem reported.
6. Check to see the Status of the problem

[Attach Screenshot]

Time	Severity	Recovery time	Status	Info	Host	Problem	Duration	Update	Actions	Tags
12:52:21 AM	Warning		PROBLEM		Windows_SERVER_Melvin	Windows: High CPU utilization (over 90% for 5m)	1m 8s	Update	class: os component: cpu scope: performance	***

Displaying 1 of 1 found

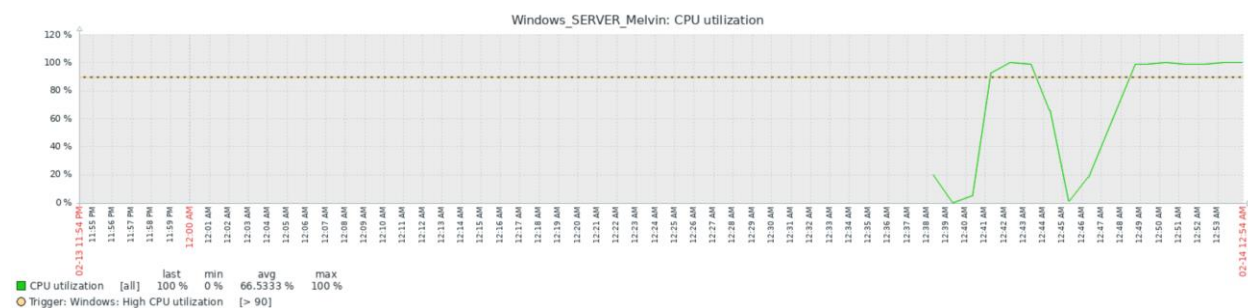
7. Click on the problem section to click the History > Physical Memory: Memory Utilization.
8. This would display a graph of the high CPU utilization usage of your windows server 2019.

Fuad Mustapha

Fuad.mustapha@durhamcollege.ca

INFT 1202

[Attach Screenshot]



9. Go over to your windows 2019 server to stop the heavy loader application
10. Once you have stopped the heavy loader software, go back to your Zabbix instance to see the status of your problems. You should see a RESOLVED status and PROBLEM status.

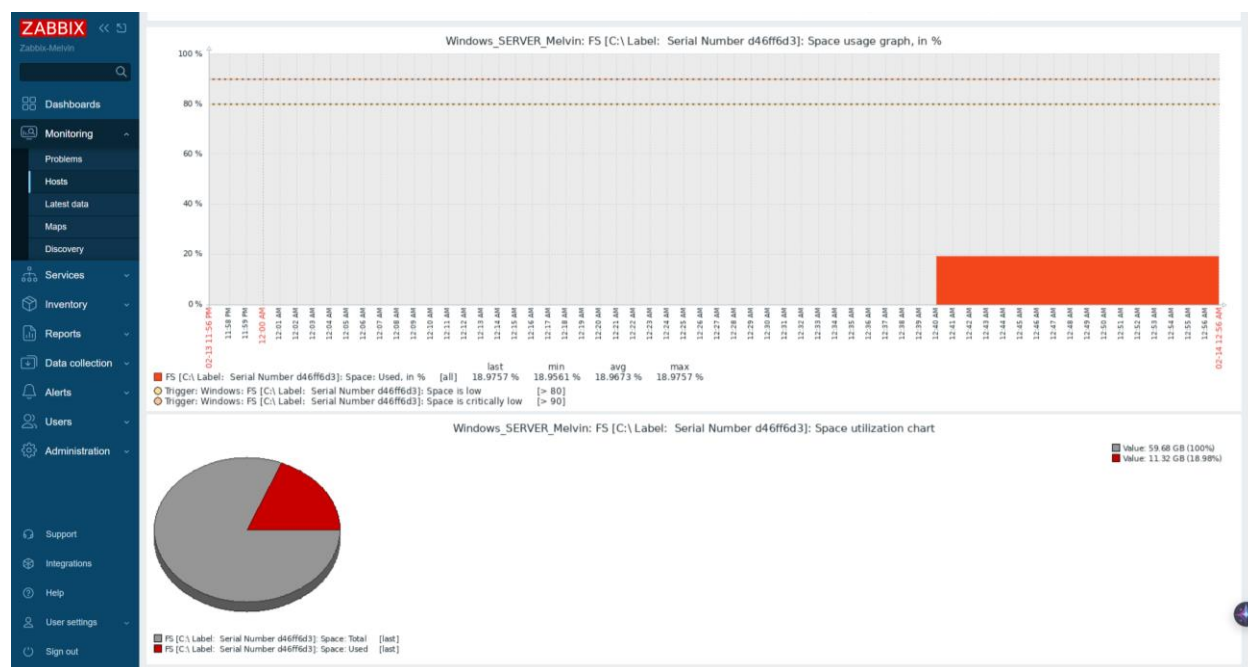
[Attach Screenshot]

☐	Time	Severity	Recovery time	Status	Info	Host	Problem	Duration	Update	Actions	Tags
☐	12:52:21 AM	Warning	12:56:21 AM	RESOLVED		Windows_SERVER_Melvin	Windows: High CPU utilization (over 90% for 5m)	4m	Update		class: os component: cpu scope: performance ***

Displaying 1 of 1 found

11. Click on the hosts from the left menu tab. Navigate to the graph row, to open the graphs.
12. The page containing all the graphs from your windows 2019 server should be displayed

[Attach Screenshot]



Congratulations, you have completed the Zabbix monitoring configuration on your Windows Server 2019.

INFT 1202

Things to Explore:

You are welcome to explore beyond the mandatory requirements if you wish.

General submission requirements

- **Marks distribution:**
 - a) 10 Screenshots (100 marks)
- **Do not alter the sequence of steps of this document. Do not delete the words [Attach the screenshot] anywhere from the assignment. Keep the numbering same. Paste the screenshots wherever it's being asked- do not use screenshots of professor's from lab manual. They are for your reference only and would be considered an academic integrity breach. You should be not even submitting another student's screenshot- Academic Integrity alert.**